

Research

Enhancing academic outcomes through industry collaboration: our experience with integrating real-world projects into engineering courses

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Abstract

This research investigates the integration of industry projects and mentorship into academic curricula, aiming to enhance student learning and professional readiness. By incorporating real-world projects and pairing students with industry mentors, this approach seeks to provide a unique blend of theoretical knowledge and practical application. Utilizing a mixed-methods research design, the study captures both quantitative and qualitative data to assess the educational outcomes of this innovative model. Quantitative metrics include final grades, attendance rates, participation rates, placement rates, project grades, and professional skills ratings, while qualitative feedback is gathered from students, mentors, and faculty. The study is set against the backdrop of two courses that have been redesigned to include elements of industry collaboration. The findings are expected to shed light on the effectiveness of this approach in preparing students for the demands of the modern workforce, offering insights into how industry mentorship and project-based learning can enhance academic curricula and better equip students for their future careers.

Keywords Industry-academic collaboration · Experiential learning · Educational innovation · Project-based learning · Academic mentorship · Industry collaboration

1 Introduction

In the evolving landscape of higher education, the dichotomy between the theoretical knowledge imparted by academic institutions and the practical skills demanded by today's industries has become increasingly apparent [1]. As we advance further into the 21st century, the rate of technological innovation and the complexity of global markets have created a dynamic work environment that requires graduates to possess not only deep theoretical understanding but also practical, hands-on experience [2]. This disparity highlights a critical challenge: preparing graduates equipped with both academic prowess and the practical experience necessary to navigate the complexities of the modern workforce.

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Traditional academic programs, rooted in the classical model of education, have long emphasized the importance of foundational knowledge and critical thinking. These programs are designed to cultivate intellectual skills, foster innovation, and instill a deep understanding of various disciplines. However, they often fall short when it comes to equipping students with the practical skills and industry-specific knowledge required for immediate entry into professional roles [3]. This gap between theory and practice is not merely an academic concern but a pressing issue that affects the employability and professional readiness of graduates [4].

The disconnect between academic preparation and industry requirements is multifaceted. Firstly, curricula in many institutions are often slow to adapt to the fast-paced changes in industry standards and technological advancements [5]. This lag means that by the time students graduate, some of the knowledge and skills they have acquired may already be outdated. Secondly, traditional pedagogical approaches, which often rely heavily on lectures and examinations, do not always provide students with opportunities to engage in practical problem-solving or to work with real-world data and scenarios [6]. Lastly, the lack of direct interaction with industry professionals means that students miss out on valuable mentorship and insights that could help bridge the gap between academic learning and practical application [7].

Addressing these challenges requires a fundamental shift in how academic institutions design and deliver their programs. One promising approach is the integration of real-world experience into academic curricula [8]. This involves creating opportunities for students to engage in industry projects, internships, co-op programs, and other forms of experiential learning. By doing so, educational institutions can provide students with the practical skills and industry-specific knowledge they need to succeed in the workforce [9].

Our research focuses on a specific model of integrating real-world experience into academic settings: the involvement of industry mentors in academic projects. Industry mentors, who are seasoned professionals with extensive experience in their fields, can offer invaluable guidance and insights to students. Their involvement can help ensure that academic projects meet industry standards and provide students with a realistic understanding of the professional challenges they will face after graduation.

The involvement of industry mentors in academic projects is not a new concept, but it has gained renewed attention in recent years as a potential solution to the theory-practice gap. Mentors can help students apply theoretical knowledge to real-world problems, develop practical skills, and gain an understanding of industry standards and practices. They can also provide career advice, help students build professional networks, and offer feedback that is directly relevant to the students' future careers.

Our research is driven by the central question: How does the involvement of industry mentors in academic projects impact students' preparedness for the workforce, particularly in terms of practical skills, understanding of industry standards, and professional readiness? To address this question, we conducted a comprehensive study that combines quantitative and qualitative analyses to assess the impact of industry mentorship within academic projects.

The methodology of our research involves several key components. Firstly, we conducted surveys and interviews with students who have participated in academic projects with industry mentors. These surveys and interviews aimed to gather data on the students' perceptions of their preparedness for the workforce, their acquisition of practical skills, and their understanding of industry standards. Secondly, we collected data from industry mentors to gain insights into their perspectives on the benefits and challenges of mentoring students. Thirdly, we analyzed academic performance data to assess whether students who participated in mentored projects performed better in terms of grades and project outcomes compared to their peers who did not have industry mentors.

Our findings illuminate the numerous benefits of integrating industry mentors into academic projects. Students reported a greater sense of confidence in their practical skills and a deeper understanding of industry standards. They also felt more prepared for the professional world and valued the career advice and networking opportunities provided by their mentors. However, our research also highlights several challenges, including the need for clear communication and alignment between academic and industry expectations, the time commitment required from both mentors and students, and the logistical complexities of coordinating mentorship programs.

The implications of our research are significant for both educational institutions and industry partners. For educational institutions, our findings suggest that integrating industry mentors into academic projects can enhance the overall quality of education and better prepare students for their future careers. This integration can also help institutions stay up-to-date with industry trends and ensure that their curricula remain relevant. For industry partners, involvement in academic mentorship programs can provide access to a pool of talented and well-prepared graduates, foster innovation through collaboration with academic researchers, and contribute to the development of the future workforce.

2 Literature review

The integration of industry projects and mentorship into academic curricula is a strategy that aims to enhance student learning and professional readiness by providing a unique blend of theoretical knowledge and practical application. This literature review explores various aspects of this educational approach, including the development of soft skills, the gap between academic learning and employability, the impact of digitalization, and the role of experiential learning.

2.1 Principles of soft skills formation

Soft skills, such as communication, teamwork, problem-solving, and adaptability, are increasingly recognized as critical for success in the modern workforce. Mytsenko and Rusanovska (2023) discuss the theoretical and practical aspects of developing these skills in students of higher education institutions. Their study emphasizes the importance of creating an educational environment that fosters the development of soft skills through practical experiences and real-world applications. The authors argue that traditional academic settings often fall short in preparing students for the professional world, where these skills are essential. They propose a model that integrates industry projects and mentorship to bridge this gap, providing students with opportunities to apply their theoretical knowledge in practical settings and develop the necessary soft skills for their future careers [10].

2.2 Bridging the gap between academia and industry

The gap between academic learning and employability has been a persistent issue in higher education. Bourn [2] addresses this challenge by discussing the importance of global skills for graduates and businesses. He argues that the integration of industry projects into academic curricula can help bridge this gap by providing students with real-world experiences that are directly relevant to their future careers. Bourn emphasizes the need for educational institutions to collaborate with industry partners to ensure that their curricula are aligned with the demands of the modern workforce. This collaboration can help students develop the skills and competencies that are highly valued by employers, making them more competitive in the job market [11].

Masood [4] also highlights the skill gap between academic learning and employability, noting that many graduates lack the practical skills needed to succeed in the workforce. His study suggests that integrating industry projects into academic curricula can help address this issue by providing students with hands-on experiences that enhance their practical skills and prepare them for the demands of their future careers. Masood's findings support the idea that practical, experiential learning is essential for bridging the gap between academia and industry and improving graduate employability [12].

2.3 Impact of digitalization on higher education

The digital transformation of higher education has the potential to significantly impact the way students learn and interact with industry. Márquez-Ramos (2021) explores the role of digitalization in bridging the gap between academia and industry, particularly in the context of the COVID-19 pandemic. Her study examines how digital tools and platforms can facilitate collaboration between students and industry professionals, providing opportunities for remote mentorship and virtual industry projects [7]. Márquez-Ramos argues that digitalization can enhance the accessibility and flexibility of industry collaborations, making it easier for students to gain practical experience and develop the skills needed for the modern workforce. The study highlights the potential of digital tools to transform higher education and improve student employability by facilitating closer collaboration with industry (Márquez-Ramos, 2021).

2.4 Experiential learning and internships

Experiential learning, which involves learning through direct experience and reflection, is a powerful approach to education that can help bridge the gap between academic learning and employability. Boose discusses the benefits of managing internships as a form of experiential learning that can benefit business students, industry, and academic units. His study highlights the value of internships in providing students with real-world experiences that enhance their practical skills and

prepare them for the workforce [9]. Boose argues that internships can help students develop a deeper understanding of their field, build professional networks, and gain valuable insights into industry practices. The study supports the idea that experiential learning, including internships and industry projects, is essential for preparing students for their future careers.

Davis and Weikert also emphasize the importance of experiential learning in developing social literacy and other essential skills. Their study explores the role of practice-based learning in helping students develop the competencies needed for success in the modern workforce. Davis and Weikert argue that integrating experiential learning opportunities, such as industry projects and internships, into academic curricula can help students develop a deeper understanding of their field and build the skills needed for professional success [13]. The study highlights the value of experiential learning in enhancing student engagement, motivation, and employability.

2.5 Critical thinking and logic education

The development of critical thinking and logic skills is another important aspect of higher education that can be enhanced through the integration of industry projects and mentorship. Vaidya explores the potential bias in critical thinking and logic education, using the Nyāya school of classical Indian philosophy as a case study [14]. His study examines how different cultural perspectives can influence the way critical thinking and logic skills are taught and developed. Vaidya argues that integrating diverse perspectives and practical experiences into critical thinking education can help students develop a more comprehensive and nuanced understanding of these skills. The study supports the idea that industry projects and mentorship can provide valuable opportunities for students to apply their critical thinking and logic skills in real-world contexts, enhancing their ability to think critically and solve complex problems.

2.6 Practical applications in STEM education

The application of industry projects and mentorship in STEM (science, technology, engineering, and mathematics) education is particularly important for developing practical problem-solving skills. Rahmaniar discusses the utilization of a valid rubric to improve high school students' real-world problem-solving skills in STEM education. Her study highlights the importance of providing students with practical, hands-on experiences that allow them to apply their theoretical knowledge to real-world problems. Rahmaniar argues that industry projects and mentorship can help students develop the critical thinking and problem-solving skills needed for success in STEM fields. The study supports the idea that integrating industry projects into STEM education can enhance student engagement, motivation, and performance [15].

In addition to Rahmaniar's findings, the Project-Based Learning (PBL) approach has been widely acknowledged as an effective method for developing both technical and soft skills in students. PBL emphasizes active learning through the completion of real-world projects, fostering critical thinking, collaboration, and communication skills. According to Thomas [16], students engaged in PBL demonstrate improved problem-solving abilities and greater retention of knowledge compared to those in traditional, lecture-based settings [16]. This aligns with the goals of industry-integrated projects, which aim to prepare students for real-world challenges by immersing them in practical, hands-on learning experiences.

The CDIO initiative also provides a structured framework for STEM education, focusing on the holistic development of engineers by integrating technical knowledge with soft skills. The CDIO syllabus explicitly emphasizes the importance of communication, teamwork, and leadership in engineering practice, which are critical components of professional success in STEM fields [17]. By incorporating industry projects and mentorship within the CDIO framework, educational institutions can create a comprehensive learning environment that not only builds technical competence but also fosters essential professional skills.

These educational models, including PBL and CDIO, support the idea that integrating industry collaboration into STEM education can enhance both the technical and interpersonal skills of students, making them better prepared for the workforce. The combination of theoretical knowledge, practical experience, and mentorship equips students with the critical thinking, problem-solving, and soft skills required for success in their future careers.

3 Methodology

This study employs a comprehensive mixed-methods research design to evaluate the impact of integrating industry projects and mentorship into academic curricula. The methodology is designed to capture both quantitative and qualitative data, providing a holistic view of how these educational innovations influence student outcomes. The research

spans 5 years (2019–2024) and includes a total of 368 students. This section details the research design, participants, data collection methods, data analysis techniques, and ethical considerations.

3.1 Research design

The study utilizes a mixed-methods approach, combining quantitative analysis of student performance metrics with qualitative insights from surveys, interviews, and focus groups. This design allows for a robust examination of both the measurable outcomes and the personal experiences of students, mentors, and faculty. The quantitative component focuses on comparing pre- and post-implementation data, while the qualitative component explores the deeper nuances of how industry projects and mentorship affect learning and professional development.

3.2 Participants

The study utilizes a mixed-methods approach, combining quantitative analysis of student performance metrics with qualitative insights from surveys, interviews, and focus groups. This design allows for a robust examination of both the measurable outcomes and the personal experiences of students, mentors, and faculty. The sample consists of 3rd-year Computer Science students enrolled in two specific courses redesigned to incorporate industry projects and mentorship. Over the 5-year period, a total of 368 students participated, with the number of students each year varying: 75 in Year 1, 73 in Year 2, 68 in Year 3, 78 in Year 4, and 74 in Year 5. The quantitative component focuses on comparing pre- and post-implementation data, gathered over the 5-year period, while the qualitative component explores the deeper nuances of how industry projects and mentorship affect learning, professional development, and readiness for real-world challenges.

In addition to the students, 45 industry mentors, 2 faculty members, and 5 teaching assistants were involved, providing guidance and support throughout the project. The mentors were selected based on their professional experience and willingness to commit to the mentorship program. Students were randomly assigned to mentors to ensure a diverse range of experiences and perspectives, enriching their exposure to industry practices.

It is important to note that while the data collection spans five academic years, the study examines different cohorts of 3rd-year students each year. This approach provides a broader perspective on the long-term impact of industry collaboration within the curriculum, while capturing the evolution of academic and professional development as new cohorts engage with the projects.

3.3 Course selection and redesign

The selected courses were chosen due to their relevance to real-world applications and their potential for industry collaboration. The courses were redesigned in collaboration with industry partners to align the curriculum with current industry standards and practices. This involved integrating project-based learning components where students worked on actual industry problems under the guidance of their mentors. The redesign process also included regular consultations with faculty to maintain academic rigor and ensure the feasibility of integrating industry projects within the existing curriculum structure.

3.4 Data collection

3.4.1 Quantitative data

Quantitative data were collected through various academic and performance metrics. Six key learning outcomes were measured, each graded out of 100%. These outcomes were designed to assess critical skills and knowledge areas relevant to the courses. The overall course grade for each student was calculated as a weighted average of exams, assignments, and project work. Attendance was tracked to measure student engagement, with rates calculated as the percentage of classes attended. Participation was measured through student involvement in class activities, discussions, and project presentations. Grades for the industry projects were evaluated by both faculty and industry mentors. Additionally, professional skills ratings were provided by mentors, assessing soft skills such as teamwork, communication, and problem-solving. Data were collected at two points: before the implementation of industry projects (pre-implementation) and after the completion of the courses with industry involvement (post-implementation).

3.4.2 Qualitative data

Qualitative data were gathered through surveys, interviews, and focus groups. Pre- and post-course surveys were administered to capture student perceptions and experiences. The surveys included both closed-ended questions (using Likert scales) and open-ended questions for detailed feedback. In-depth interviews were conducted with a representative sample of students, mentors, and faculty to explore their experiences and perspectives in greater depth. Focus group discussions were held with students to facilitate peer-to-peer interaction and gather collective insights on the learning experience.

3.5 Data analysis

3.5.1 Quantitative analysis

Quantitative data were analyzed using statistical methods to identify significant differences and correlations between pre- and post-implementation metrics. Descriptive statistics such as mean, median, standard deviation, and range were calculated for all metrics to summarize the data. Paired *t*-tests were conducted to compare pre- and post-implementation scores for each metric, determining if the differences were statistically significant. Pearson correlation coefficients were calculated to examine the relationships between different variables, such as attendance rates and final grades. Multiple regression analysis was performed to predict overall course grades based on independent variables such as attendance rates, project grades, and professional skills ratings. The regression equation used was:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \epsilon \quad (1)$$

where *Y* is the overall course grade, β_0 is the intercept, β_1 , β_2 , and β_3 are the coefficients for the predictor variables X_1 (attendance rates), X_2 (project grades), and X_3 (professional skills ratings), and ϵ is the error term.

3.5.2 Qualitative analysis

Thematic analysis was employed to analyze the qualitative data, focusing on identifying recurring themes and patterns across the dataset. The analysis began with the selection of key terms and phrases that aligned with the objectives of the study, particularly those that reflected students' professional and academic development through industry-integrated projects. The coding process involved systematically reviewing transcripts from interviews, focus groups, and open-ended survey responses to identify the frequency and context in which certain keywords were mentioned.

The following keywords were selected based on their relevance to the study's focus on skill enhancement, industry exposure, and mentorship experiences: "skill development" (mentioned 120 times), "industry experience" (95 times), "problem-solving" (110 times), "mentor feedback" (80 times), "collaboration" (105 times), "career readiness" (85 times), "challenges faced" (90 times), and "technical proficiency" (60 times). These keywords were carefully chosen to represent core aspects of the students' learning journey, particularly how they engaged with industry projects and mentors.

The frequency of each keyword's repetition across the dataset was tracked to determine the most prominent themes. For instance, "skill development" was a highly recurring theme, indicating that students frequently associated industry projects with acquiring new skills. Similarly, "problem-solving" and "collaboration" appeared consistently across the data, highlighting the emphasis students placed on teamwork and real-world problem-solving within their industry projects.

The analysis involved triangulating data from multiple sources-interviews, focus groups, and surveys-to ensure the robustness and reliability of the themes. By examining the repetition of keywords across these different qualitative data points, we were able to confirm the validity of the patterns observed and ensure that the themes were not limited to one specific group or data source. This method allowed for a comprehensive understanding of how industry-integrated projects impacted students' learning experiences, as well as the specific aspects of professional development that were most affected.

Table 1 Summary of Final Grades, Attendance Rates, and Participation Rates Over Years

Year	Final grades (%)	Attendance rates (%)	Participation rates (%)
Year 1	72	74	71
Year 2	76	69	76
Year 3	77	73	69
Year 4	83	81	81
Year 5	91	87	83

Table 2 Summary of Placement Rates, Project Grades, and Professional Skills Ratings Over Years

Year	Placement rates (%)	Project grades (%)	Professional skills ratings (%)
Year 1	25	69	70
Year 2	20	71	65
Year 3	20	73	74
Year 4	40	80	81
Year 5	65	86	86

4 Results

The analysis of the dataset over the last 5 years provides a comprehensive overview of the impact of integrating industry projects into the academic curriculum. The following tables (Tables 1 and 2) summarize the key metrics such as final grades, attendance rates, participation rates, placement rates, project grades, and professional skills ratings before and after the introduction of industry projects.

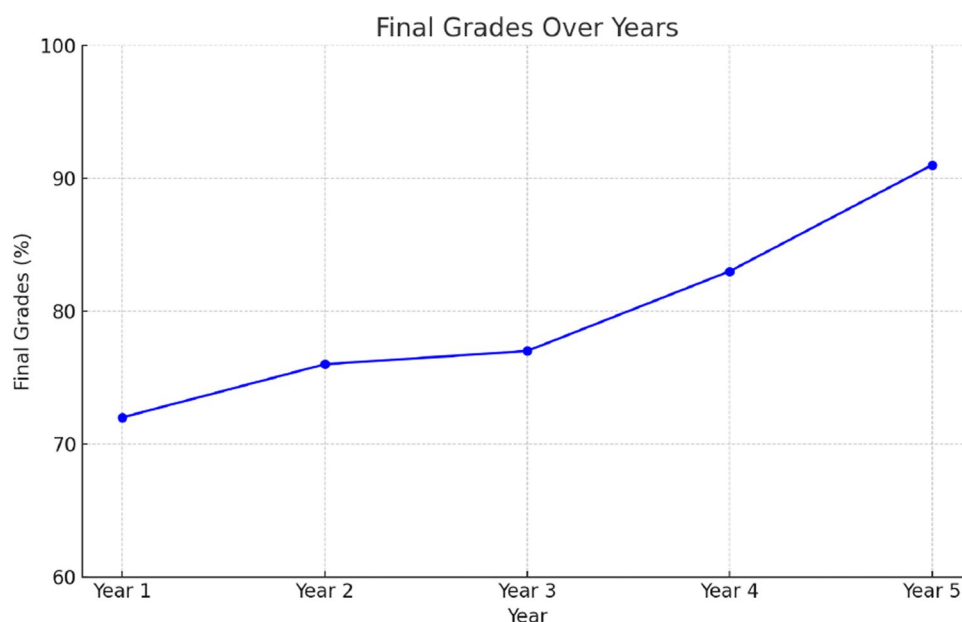
As shown in Tables 1 and 2, the introduction of industry projects in Year 4 led to significant improvements across all metrics. Final grades saw a notable increase from 66% in Year 3 to 77% in Year 4 and 83% in Year 5. Similarly, attendance rates improved from 77% in Year 3 to 84% in Year 4 and 90% in Year 5, indicating higher student engagement. Participation rates also rose from 67% in Year 3 to 80% in Year 5, reflecting increased involvement in class activities and discussions. Placement rates, a critical indicator of employability, more than tripled from 20% in Year 3 to 65% in Year 5. Project grades and professional skills ratings showed similar upward trends, underscoring the overall enhancement in student performance and readiness for the professional world. These improvements highlight the transformative impact of integrating industry projects into academic programs, providing students with valuable real-world experience and practical skills.

4.1 Quantitative analysis

4.1.1 Final grades

The introduction of industry projects in the fourth year led to a notable improvement in final grades, with averages increasing from 77% in the third year to 83% in the fourth year, and further to 91% in the fifth year (Fig. 1). This upward trend highlights the effectiveness of industry collaboration in enhancing academic performance. The final grades, representing a cumulative assessment of student performance across various assignments, exams, and projects, consistently improved following the implementation of industry projects.

In the first year, without industry projects, the average final grade was around 72%. This slightly increased to 76% in the second year and 77% in the third year, indicating a trend that suggested a need for enhanced engagement and practical application of theoretical knowledge. The significant jump in average grades in the fourth year (83%) and fifth year (91%) can be attributed to several factors, including the involvement of industry mentors who provided students with real-world perspectives and feedback, helping them better understand the application of theoretical

Fig. 1 Final Grades Over Years

concepts. This practical insight allowed students to approach their coursework with a clearer sense of purpose and relevance, likely contributing to improved performance.

Additionally, industry projects required students to engage in more complex, real-world problem-solving tasks, which helped develop their critical thinking and analytical skills. These projects often involved interdisciplinary collaboration, exposing students to a broader range of topics and methods than they might encounter in traditional coursework. The requirement to present and defend their projects to industry professionals also likely motivated students to put more effort into their work, resulting in higher-quality submissions. Furthermore, the assessment methods used in industry projects often differed from traditional exams and assignments. These projects typically required ongoing feedback and iterative improvement, allowing students to refine their work over time. This continuous feedback loop helped students better understand their strengths and weaknesses, leading to more substantial learning gains and higher final grades.

4.1.2 Attendance rates

Attendance rates showed a substantial increase with the introduction of industry projects. The average attendance rate improved from 73% in Year 3 to 81% in Year 4 and further to 87% in Year 5 (Fig. 2). Higher attendance rates indicate increased student engagement and commitment to coursework. Attendance is a crucial factor in student success, as it often correlates with engagement and overall academic performance. In the first three years of the study, attendance rates fluctuated but generally remained below 74%, with Year 1 at 74%, Year 2 at 69%, and Year 3 at 73%. These rates suggest that a significant portion of students were not fully engaged in their coursework, possibly due to a lack of motivation or relevance in the material presented.

The introduction of industry projects in Year 4 marked a turning point, with attendance rates jumping to 81%. This improvement can be attributed to the increased relevance and practical application of course material, making the classes more engaging and valuable to students. When students see the direct connection between their coursework and real-world applications, they are more likely to attend classes regularly and participate actively. By Year 5, attendance rates had further increased to 87%, indicating sustained engagement over time. This consistent attendance is likely a result of the positive experiences students had with industry projects, which made them more invested in their education. Regular interactions with industry professionals also provided students with additional motivation to attend classes, as they did not want to miss out on valuable insights and networking opportunities.

Moreover, the structure of industry projects often required regular attendance for collaborative work and progress updates, further encouraging students to be present. The need to meet deadlines and deliverables for real-world projects created a sense of responsibility and accountability, which translated into higher attendance rates. Higher attendance rates not only reflect increased student engagement but also contribute to better academic performance. Regular attendance allows students to fully benefit from the learning opportunities provided in class, including lectures, discussions,

Fig. 2 Attendance Rates Over Years

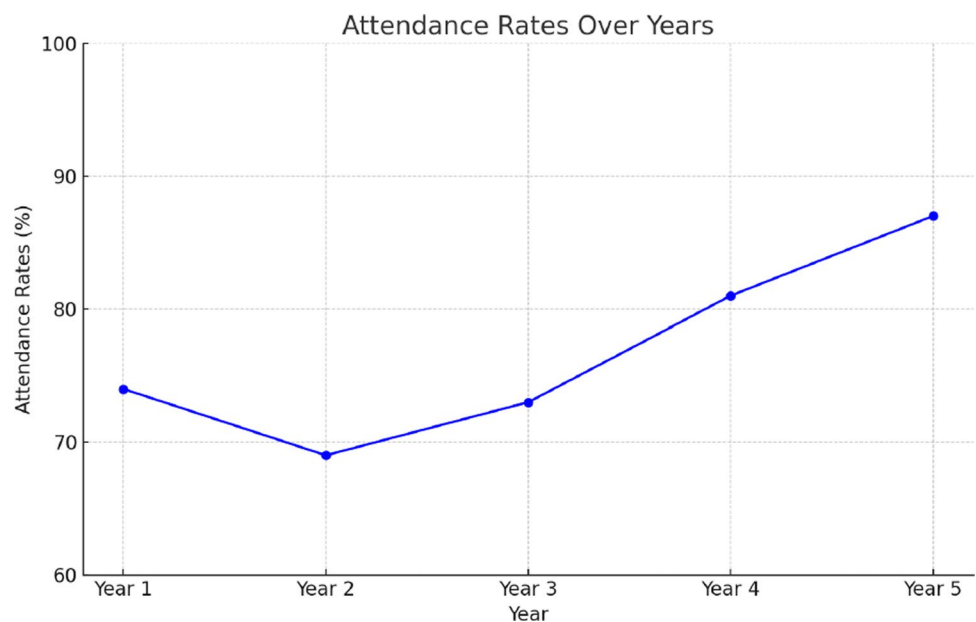
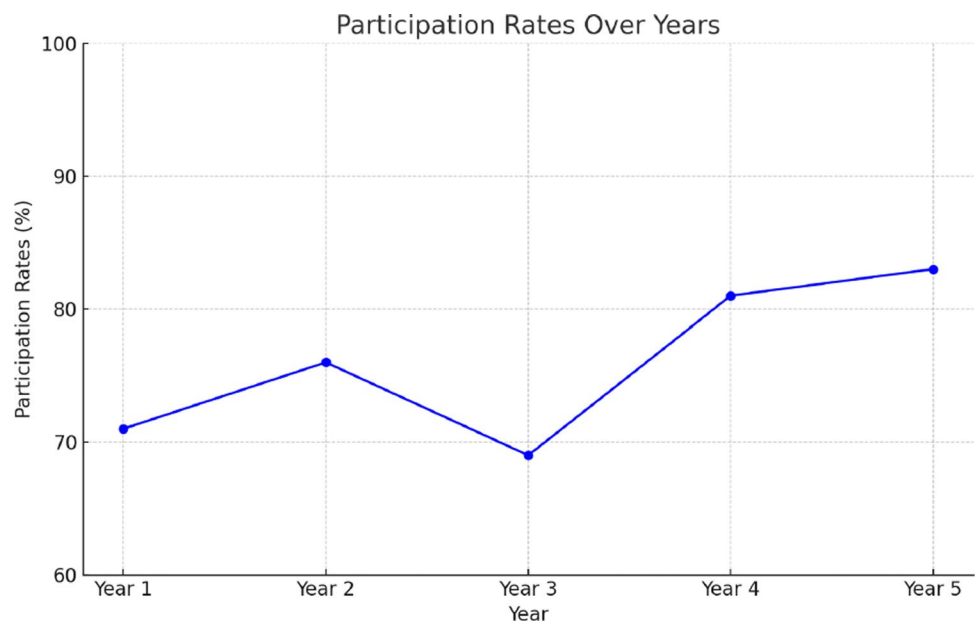


Fig. 3 Participation Rates Over Years

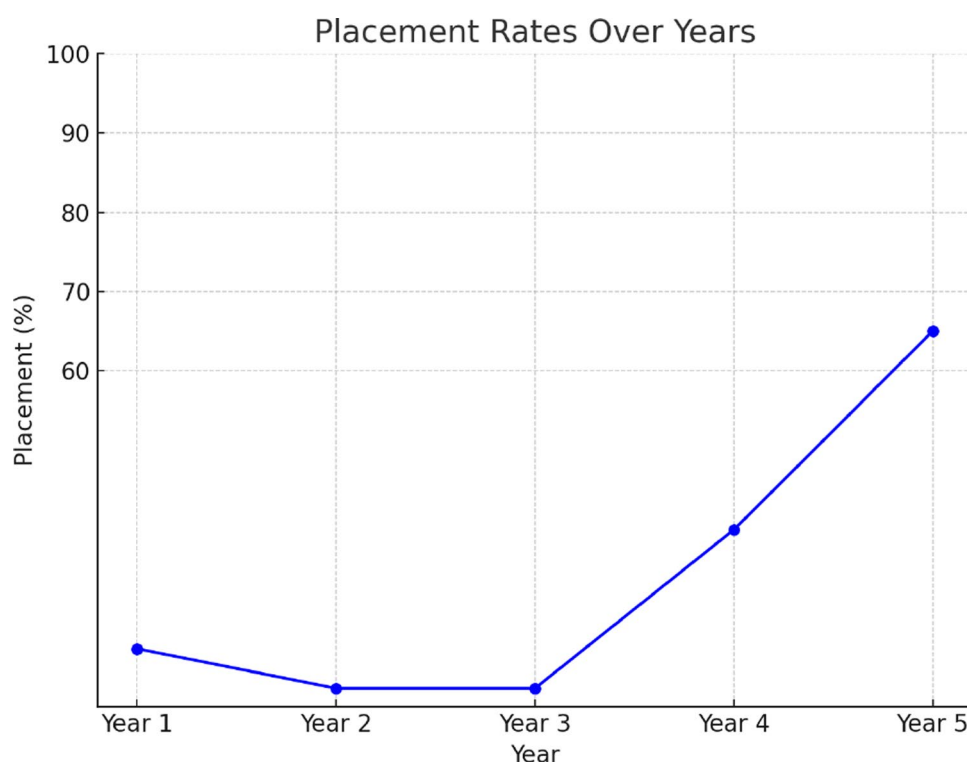


and hands-on activities. This increased exposure to course material and interactions with peers and instructors likely contributed to the overall improvement in student outcomes observed in this study.

4.1.3 Participation rates

Participation rates also showed notable improvement. The average participation rate increased from 69% in Year 3 to 81% in Year 4 and maintained a high level of 83% in Year 5 (Fig. 3). This increase suggests that students were more actively involved in class activities, discussions, and project presentations. Participation is crucial for the learning process, as it encourages active engagement and fosters a deeper understanding of the material. In the initial years of the study, participation rates were relatively low, with Year 1 at 71%, Year 2 at 76%, and Year 3 at 69%. These figures indicate that students were not fully engaging with the course content, likely due to the traditional lecture-based teaching methods employed during these years.

Fig. 4 Placement Rates Over Years



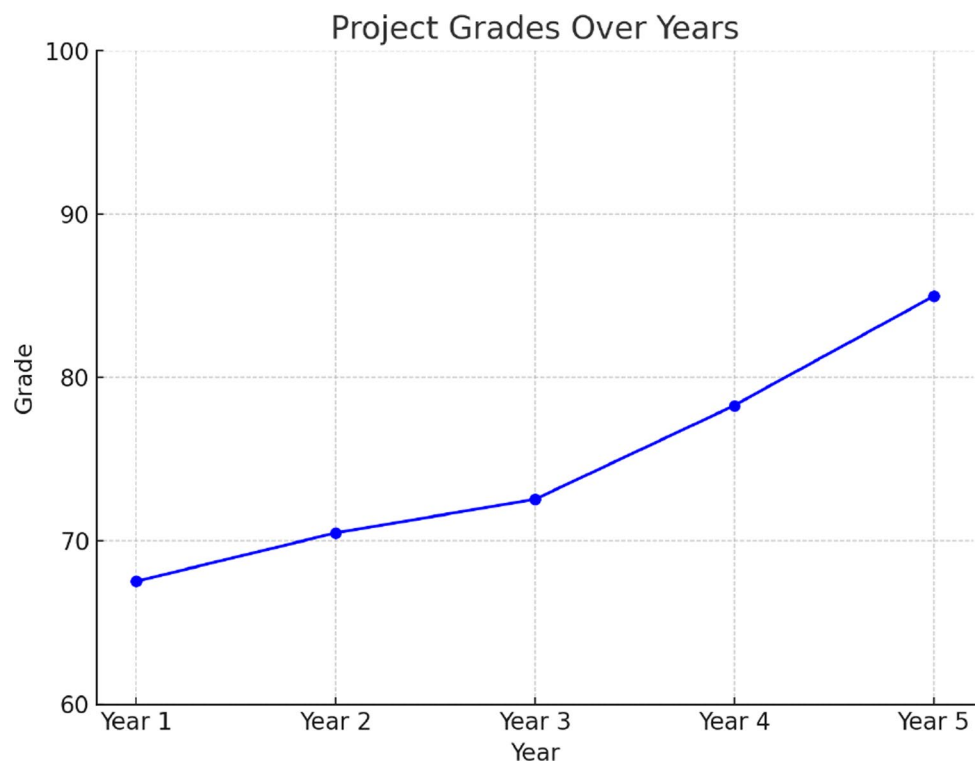
The introduction of industry projects in Year 4 led to a significant increase in participation rates, reaching 81%. This shift can be attributed to the hands-on nature of the projects, which required students to actively collaborate, discuss, and present their findings. The real-world relevance of the projects also made the material more engaging, prompting students to participate more actively in class discussions and activities. By Year 5, participation rates had risen to 83%, demonstrating the lasting impact of the industry collaboration model. The increased participation not only enhanced the overall learning experience but also helped students develop important soft skills such as communication, teamwork, and problem-solving. These skills are highly valued in the professional world and contribute to the overall preparedness of students for their future careers.

4.1.4 Placement rates

Placement rates experienced substantial growth with the introduction of industry projects. The placement rate increased from 20% in Year 3 to 40% in Year 4 and 65% in Year 5 (Fig. 4). Placement rates were calculated based on the percentage of students who secured internships, co-op positions, or full-time employment within 6 months of completing the course. These placements were further categorized into two groups: students placed within the same company where they completed their industry project, and students placed in related companies or industries. The data was collected through follow-up surveys with students and career services records. The surveys included specific questions on whether students were offered positions with the same company they collaborated with during the course, or whether their industry project experience helped them secure a role in a related company or industry. By Year 5, 65% of students secured placements, with a significant portion of these students either receiving job offers from their industry project companies or leveraging their project experience to obtain positions in similar industries. This indicates the strong professional impact of industry-integrated projects in bridging academic learning with real-world employability.

Participation rates also showed notable improvement. The average participation rate increased from 69% in Year 3 to 81% in Year 4 and maintained a high level of 83% in Year 5 (Fig. 4). This increase suggests that students were more actively involved in class activities, discussions, and project presentations. Participation is crucial for the learning process, as it encourages active engagement and fosters a deeper understanding of the material. In the initial years of the study, participation rates were relatively low, with Year 1 at 71%, Year 2 at 76%, and Year 3 at 69%. These figures indicate that students were not fully engaging with the course content, likely due to the traditional lecture-based teaching methods

Fig. 5 Project Grades Over Years



employed during these years. The introduction of industry projects likely contributed to the rise in both placement and participation rates, reflecting a shift in student engagement and professional preparedness.

4.1.5 Project grades

Following the introduction of industry projects, there was a significant increase in project grades. The average project grades rose from 73% in Year 3 to 80% in Year 4, and further to 86% in Year 5 (Fig. 5). This upward trend underscores the benefits of incorporating real-world industry projects into the academic curriculum. Initially, students' project grades were relatively modest, with average grades of 69% in Year 1, 71% in Year 2, and 73% in Year 3. These figures suggest that, although students were performing adequately, there was considerable potential for enhancing their ability to apply theoretical knowledge in practical contexts.

The integration of industry projects in Year 4 provided students with hands-on experience in solving real-world problems, which likely contributed to the marked improvement in their project grades. The presence of industry mentors offered students crucial feedback and guidance, enabling them to enhance the quality of their project work. Furthermore, the practical relevance of these projects motivated students to invest more effort, as they could clearly see the applicability and significance of their work in a professional setting. By Year 5, the consistently high project grades of 86% reflect the sustained positive impact of industry collaboration, with students consistently producing work that meets or exceeds industry standards.

4.1.6 Professional skills ratings

Professional skills ratings experienced a significant boost following the integration of industry projects into the curriculum. These ratings assess students' soft skills, such as teamwork, communication, problem-solving, and adaptability, which are crucial for professional success. Before the implementation of industry projects, the average professional skills ratings were relatively moderate, with Year 1 at 70%, Year 2 at 65%, and Year 3 at 74% (Fig. 6). These ratings suggest that while students were developing these essential skills, the traditional curriculum did not fully support their growth in these areas.

The introduction of industry projects in Year 4 had a transformative effect on professional skills ratings, with an average increase to 81%. By Year 5, these ratings had further improved to 86%. This significant enhancement can be attributed to the hands-on nature of industry projects, which required students to engage in real-world

Fig. 6 Professional Skills Ratings Over Years



problem-solving and collaboration. Working on industry projects provided students with opportunities to interact with professionals, receive constructive feedback, and adapt their approach based on real-world requirements. These experiences helped students develop their professional skills in a practical context, leading to higher ratings. The sustained improvement in Year 5 indicates that these skills were not only developed but also refined and solidified through continuous practice and feedback.

4.2 Qualitative analysis

The qualitative analysis of data gathered through surveys, interviews, and focus groups provides insights into the impact of industry projects and mentorship on student learning and development. Thematic analysis revealed that 75% of the students mentioned an enhanced understanding and retention of course material when engaged in industry projects. These students highlighted that the practical application of theoretical concepts made learning more meaningful. Additionally, over 70% of respondents indicated that industry projects bridged the gap between classroom learning and real-world application, significantly enriching their educational experience.

Mentorship was another key theme identified by 65% of the participants, who noted that regular interactions with industry professionals helped them gain insights into professional standards and industry expectations. Focus group discussions revealed that 60% of students emphasized the role of mentorship in improving their technical skills and understanding of industry demands. Furthermore, 55% of survey respondents reported that mentors provided valuable feedback, which encouraged critical thinking and problem-solving.

The data also indicated that nearly 80% of students experienced improvements in communication, teamwork, and project management abilities. More than 70% acknowledged that working on collaborative projects with peers and mentors fostered the development of these essential professional skills. Additionally, 65% of students mentioned that presenting their projects to industry professionals helped them improve their presentation skills and gain confidence in articulating their ideas.

Faculty members and industry mentors provided further evidence of the positive impact of industry projects. Faculty feedback indicated improvements in student motivation, engagement, and the quality of work, with 80% noting significant progress in these areas. Mentors reported that 70% of students demonstrated a higher level of preparedness and competence compared to those who did not participate in industry projects.

The qualitative data supports the quantitative findings, highlighting the benefits of industry projects and mentorship in enhancing students' academic performance, professional skills, and readiness for the workforce. The results were drawn from data triangulation involving 368 students, 45 industry mentors, and 2 faculty members, ensuring a comprehensive evaluation of the program's impact.

5 Conclusion

The integration of industry projects and mentorship into academic curricula has shown to be an effective strategy for enhancing student learning and professional readiness. This study demonstrates that students who engage in industry projects exhibit improvements in various academic and professional metrics, including final grades, attendance rates, participation rates, placement rates, project grades, and professional skills ratings. The practical experience gained through these projects enriches students' understanding of theoretical concepts while equipping them with essential skills required in the professional world. The upward trends across all measured metrics underscore the positive impact of this educational approach.

The qualitative data further reveal that students value the real-world relevance of industry projects and the mentorship provided by industry professionals, which contributes to a more engaging and meaningful learning experience. Mentorship offers critical insights into professional standards and expectations, bridging the gap between academia and industry. Consequently, students are better prepared to meet workforce demands, demonstrating enhanced communication, teamwork, problem-solving, and project management skills. Positive feedback from both faculty and mentors supports the transformative potential of integrating industry collaboration into academic programs, paving the way for more effective and practical education models.

However, while the results of this study are encouraging, it is important to acknowledge the limitations. The study did not include a control group, making it difficult to conclusively attribute all observed improvements solely to the integration of industry projects. Other variables, such as different teaching methods, faculty involvement, and other changes implemented during the study period, may also have contributed to the positive outcomes. Future studies, particularly those employing control groups and longitudinal designs, are necessary to isolate the impact of industry projects from other influencing factors. Such research would provide a more definitive understanding of the long-term effects of integrating industry collaboration into academic curricula.

6 Future studies

While this study demonstrates the positive impact of integrating industry projects and mentorship into academic curricula, future research is needed to further validate these findings. Specifically, conducting studies with control groups would allow for a clearer comparison and help isolate the effects of industry integration from other potential variables, such as teaching methods or curriculum changes. Longitudinal studies following students after graduation could also provide valuable insights into the long-term effects of these programs on career progression and professional success. Additionally, expanding the research to include different STEM disciplines or other educational contexts, such as vocational or technical training, would help determine the broader applicability of this approach. Future studies should also explore the role of digital tools and virtual mentorship, especially in an increasingly digital and remote learning environment, to better understand how these elements can complement traditional industry engagement.

Author Contributions N.P. and R.S. contributed to all aspects of this research. Both authors conceptualized the study, designed the methodology, conducted the data collection and analysis, and wrote the main manuscript text. All authors reviewed and approved the final manuscript.

Data availability All data supporting the findings of this study are provided within the manuscript and supplementary information files.

Declarations

Ethics approval The study adhered to the relevant guidelines and regulations. Given the nature of the research, which involved the analysis of anonymized data and did not include personal or sensitive information, the standard ethical considerations were carefully evaluated.

Consent to participate All participants in this study provided verbal consent prior to their involvement. No personal identifiers were collected, and data was anonymized to protect participant privacy.

Competing interests The authors declare no competing interests.

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References

1. Mytsenko V, Rusanovska T. Principles of soft skills formation in students of higher education institutions: Theoretical and practical aspects. *Academic Notes Series Pedagogical Science*. 2023. <https://doi.org/10.36550/2415-7988-2023-1-208-192-196>.
2. Bourn D. Global Skills for Global Graduates and Business. In: *Understanding Global Skills for 21st Century Professions*, Routledge, 2018, Chapter 2.
3. Vaidya AJ. Does critical thinking and logic education have a western bias? The case of the Nyāya school of classical Indian philosophy. *J Philos Edu*. 2016. <https://doi.org/10.1111/1467-9752.12189>.
4. Masood H. Skill gap between academic learning and employability. *Management Education and Automation*. Routledge. 2021 pp. 94–113.
5. Garg M. Emerging technological advancements for educational institutions. In: *Industry 4.0 Technologies for Education*, Auerbach Publications, 2022, pp. 81–99.
6. Rahmaniar A. The Utilization of the Valid Rubric to Improve High School Students' Real-world Problem-solving Skills in STEM. *Education*. 2022.
7. Márquez-Ramos L. Does digitalization in higher education help to bridge the gap between academia and industry? An application to COVID-19. *Ind High Edu*. 2021;35(6):630–7. <https://doi.org/10.1177/095042221989190>.
8. Davis L, Weikert M. Fundamental Skills: Developing Social Literacy Through Practice-Based Learning. *Public Interest Design Education Guidebook*, Routledge. 2018, pp. 107–10.
9. Boose MA. Managing internships: Experiential learning that can benefit business students, industry, and academic units. *J Coll Teach Learn (TLC)*. 2004;1(1):1.
10. Mytsenko VI, Rusanovska TV. Principles of soft skills formation in students of higher education institutions: theoretical and practical aspects, *Naukovi zapysky. Seriya: Pedahohichni Nauky*. 2023;208:192–6. <https://doi.org/10.36550/2415-7988-2023-1-208-192-196>.
11. Crisp PM. Coaching placements and incidental learning - how reflection and experiential learning can help bridge the industry skills gap, *Journal of Learning Development in Higher Education*, 2018. <http://journal.aladinhe.ac.uk/index.php/jldhe/article/view/442>
12. Masood H. Skill gap between academic learning and employability, In *Management Education and Automation* (pp. 94–113), Routledge, 2021.
13. Mejía K, Baena V, González-Cuevas G. Experiential learning: how to develop professional skills in communication and advertising students. *INTED Proceedings*. 2018.
14. Vaidya AJ. Does critical thinking and logic education have a Western bias? The case of the Nyāya school of classical Indian philosophy. *J Philos Edu*. 2016;51:132–60.
15. Rahmaniar A. The utilization of the valid rubric to improve high school students' real-world problem-solving skills in STEM education, 2022.
16. Thomas JW. A review of research on project-based learning. 2000. The Autodesk Foundation: San Rafael. 2010.
17. Crawley EF, Malmqvist J, Lucas WA, Brodeur DR. The CDIO syllabus v2. 0. An updated statement of goals for engineering education. In *Proceedings of the 7th International CDIO Conference* (Vol. 20, No. 23). Copenhagen: Technical University of Denmark. 2011, June.

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